Exercise 10

Create a data visualization (e.g., pie charts, bar graphs) for an inventory management system using JavaScript

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AIM:

The aim is to create data visualizations, such as pie charts and bar graphs, for an inventory management system using JavaScript.

PROCEDURE:

Step 1: Set Up Your HTML File

First, create an HTML file to hold your canvas for the charts and include Chart.js via CDN.

index.html

```
<!DOCTYPE html> <html lang="en"> <head> <meta charset="UTF-8"> <meta name="viewport" content="width=device-width, initial-scale=1.0"> <title>Inventory Management Visualization</title> <style> body { font-family: Arial, sans-serif; text-align: center; margin: 50px; } canvas { margin: 20px auto; } </style> </head> <body> <h1>Inventory Management System</h1> <canvas id="pieChart" width="400" height="400"></canvas> <canvas id="barChart" width="400" height="400"></canvas> <script src="https://cdn.jsdelivr.net/npm/chart.js"></script> <script src="script.js"></script> </body> </html>
```

Step 2: Create the JavaScript File for Charts

Next, create a JavaScript file (script.js) to handle the data visualization logic. The same inventoryData object is shared between both the pie chart and the bar chart.

script.js

```
// Data for inventory const inventoryData = { labels: ['Electronics', 'Clothing', 'Home Appliances', 'Books', 'Toys'], datasets: [{ label: 'Items in Stock', data: [200, 150, 100, 80, 50], backgroundColor: [ '#FF6384', '#36A2EB', '#FFCE56', '#4BC0C0', '#9966FF' ] }] }; // PIE Chart const ctxPie = document.getElementById('pieChart') .getContext('2d'); new Chart(ctxPie, { type: 'pie', data: inventoryData, options: { responsive: true, plugins: { title: { display: true, text: 'Inventory Distribution' } } } }); // BAR Chart const ctxBar = document.getElementById('barChart') .getContext('2d'); new Chart(ctxBar, { type: 'bar', data: inventoryData, options: { responsive: true, plugins: { title: { display: true, text: 'Items in Stock by Category' } }, scales: { y: { beginAtZero: true } } } });
```

OUTPUT:

Opening index.html in a browser renders the page with two charts displayed vertically — a Pie Chart showing inventory distribution by category, and a Bar Chart showing items in stock per category.

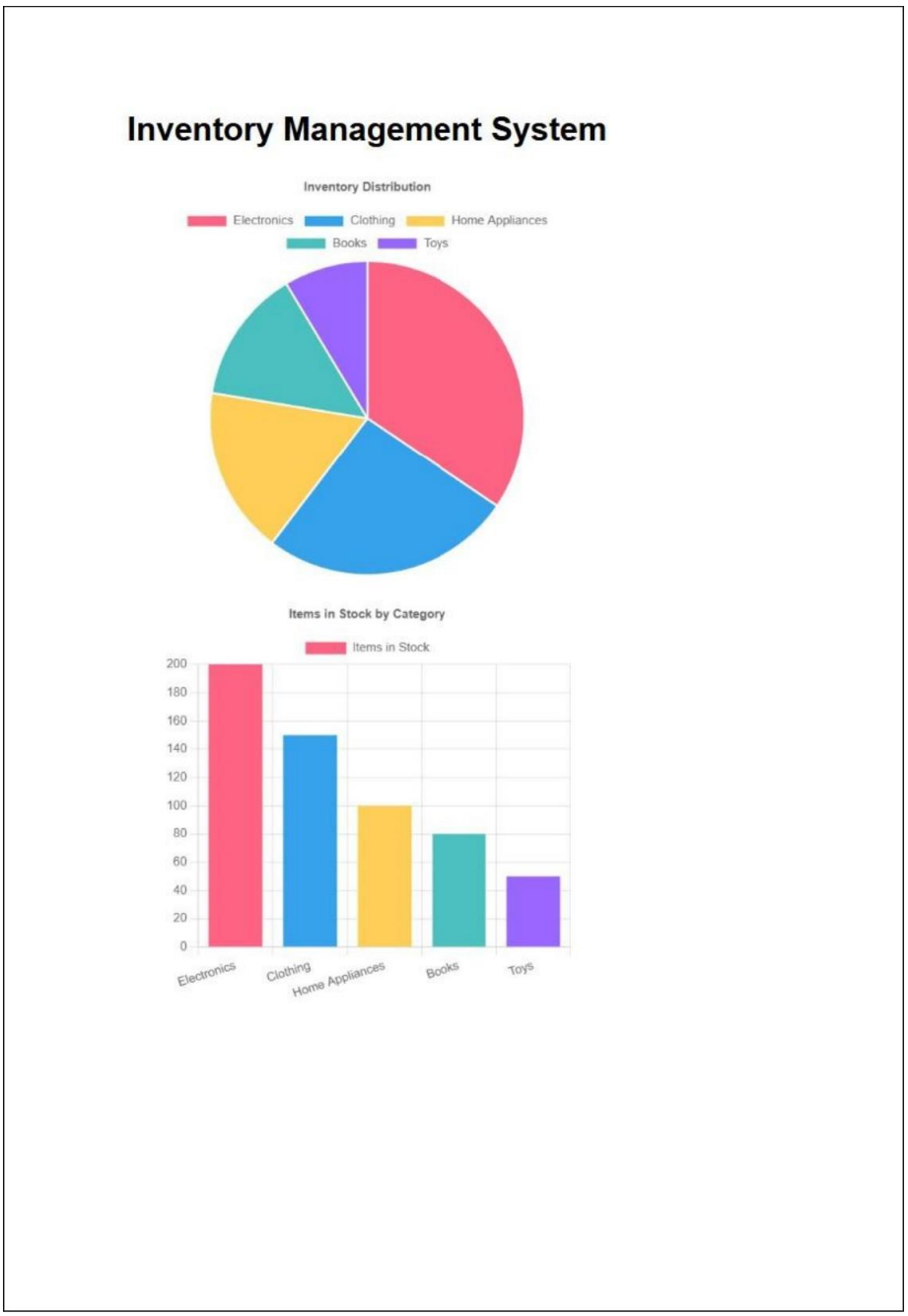

Fig: Pie chart (Inventory Distribution) and Bar chart (Items in Stock by Category)